Resume — Electrical & Quantum Engineer (Test)

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Dr. Jordan Reyes-Tanaka

Electrical & Quantum Hardware Engineer · Superconducting qubits · Cryogenic RF control electronics

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Fictional candidate for testing. All names, employers, figures and links are invented.

Summary

Electrical engineer with 7 years building control and readout hardware for superconducting quantum processors. Led the design of a 64-channel cryogenic RF control stack that cut wiring heat load at 4 K by 38% and raised two-qubit gate fidelity from 99.1% to 99.6%. Comfortable across the full chain: PCB and RF design, FPGA firmware, dilution-refrigerator integration and Python calibration automation. Seeking a senior hardware role scaling qubit systems past 1,000 physical qubits.

Technical Skills

Area	Skills
Quantum hardware	Transmon & fluxonium qubits, flux-tunable couplers, dispersive readout, JPA/TWPA amplifiers, randomized benchmarking, T1/T2 characterization
RF & microwave	IQ mixing, LO distribution, 4–8 GHz filter design, S-parameter analysis, VNA/spectrum analyzer, noise-figure budgeting

Area	Skills
Cryogenics	Bluefors & Oxford dilution refrigerators, cryo-CMOS, thermal budgeting at 4 K / 100 mK / 10 mK, coax & flex wiring
Electronics design	Altium Designer, Cadence Virtuoso, 12-layer mixed-signal PCB, SPICE, high-speed DAC/ADC (2–6 GS/s)
Firmware & FPGA	Verilog, VHDL, Xilinx RFSoC, Vivado, real-time feedback under 400 ns
Software	Python, Qiskit, QCoDeS, Labber, NumPy/SciPy, C++, MATLAB, Git, CI for lab automation
Simulation	Ansys HFSS, Sonnet, COMSOL, Keysight ADS, scqubits

Experience

Senior Quantum Hardware Engineer — Coherent Lattice Systems (fictional), Boulder, CO · Mar 2022 – Present

- Led a team of 5 designing a 64-channel cryogenic RF control stack for a 72-qubit superconducting processor.
- Cut 4 K heat load by 38% by replacing semi-rigid coax with custom NbTi flex cables and on-chip attenuation.
- Raised median two-qubit CZ gate fidelity from 99.1% to 99.6% through better flux-line filtering and pulse shaping.
- Built an automated Python calibration pipeline that shrank daily tune-up from 6 hours to 45 minutes.
- Co-inventor on 2 pending patents in multiplexed qubit readout.

RF / Mixed-Signal Engineer — Northfield Quantum Labs (fictional), Cambridge, MA · Jul 2019 – Feb 2022

- Designed a 16-channel RFSoC-based arbitrary waveform system (6 GS/s, 14-bit) for qubit drive and readout.
- Implemented FPGA active-reset feedback with 380 ns latency, raising experiment repetition rate 4×.

- Designed 4–8 GHz band-pass and Purcell filters in HFSS; matched simulation to measurement within 2%.
- Wrote test procedures that cut board bring-up from 3 weeks to 5 days.

Electrical Engineering Intern — Vantage Power Electronics (fictional), Austin, TX · May 2018 – Aug 2018

- Designed a 48 V to 12 V, 500 W synchronous buck converter reaching 96.4% peak efficiency.
- Built a LabVIEW test rig that automated thermal and load-step testing for 20 prototype boards.

Selected Projects

- **Frequency-multiplexed readout (8 qubits per feedline)** — Designed a resonator and filter chain reaching 98.7% single-shot readout fidelity in 600 ns.
- **Cryo-CMOS pulse generator prototype** — Taped out a 28 nm test chip running at 4 K under 2 mW per channel, replacing room-temperature DACs.
- **Open-source qubit tune-up toolkit** — Python library for automated spectroscopy, Rabi and Ramsey fits; 600+ GitHub stars (fictional).
- **Senior capstone: superconducting resonator Q-factor study** — Fabricated aluminum CPW resonators with internal Q above 1×10^6.

Education

Degree	School	Years	Notes
Ph.D., Electrical Engineering (Quantum Devices)	Lakeshore Institute of Technology (fictional)	2015 – 2019	Thesis: Low-loss microwave packaging for superconducting qubits
B.S., Electrical Engineering, Physics minor	University of Westbrook (fictional)	2011 – 2015	GPA 3.86 / 4.00, summa cum laude

Publications, Patents & Certifications

- J. Reyes-Tanaka et al., "Scalable flex-cable wiring for cryogenic qubit control," *Journal of Applied Quantum Engineering* (fictional), 2024.
- J. Reyes-Tanaka, M. Ode, "Purcell-filtered multiplexed readout at 98.7% fidelity," *IEEE Quantum Week* (fictional paper), 2023.
- Patent pending: "Frequency-multiplexed readout with tunable Purcell filtering," US App. 00/000,001 (fictional), 2025.
- Patent pending: "Low-heat-load cryogenic signal distribution," US App. 00/000,002 (fictional), 2025.
- Certifications: IPC CID+ (PCB design), Xilinx RFSoC Design Specialist, Professional Engineer (EIT).